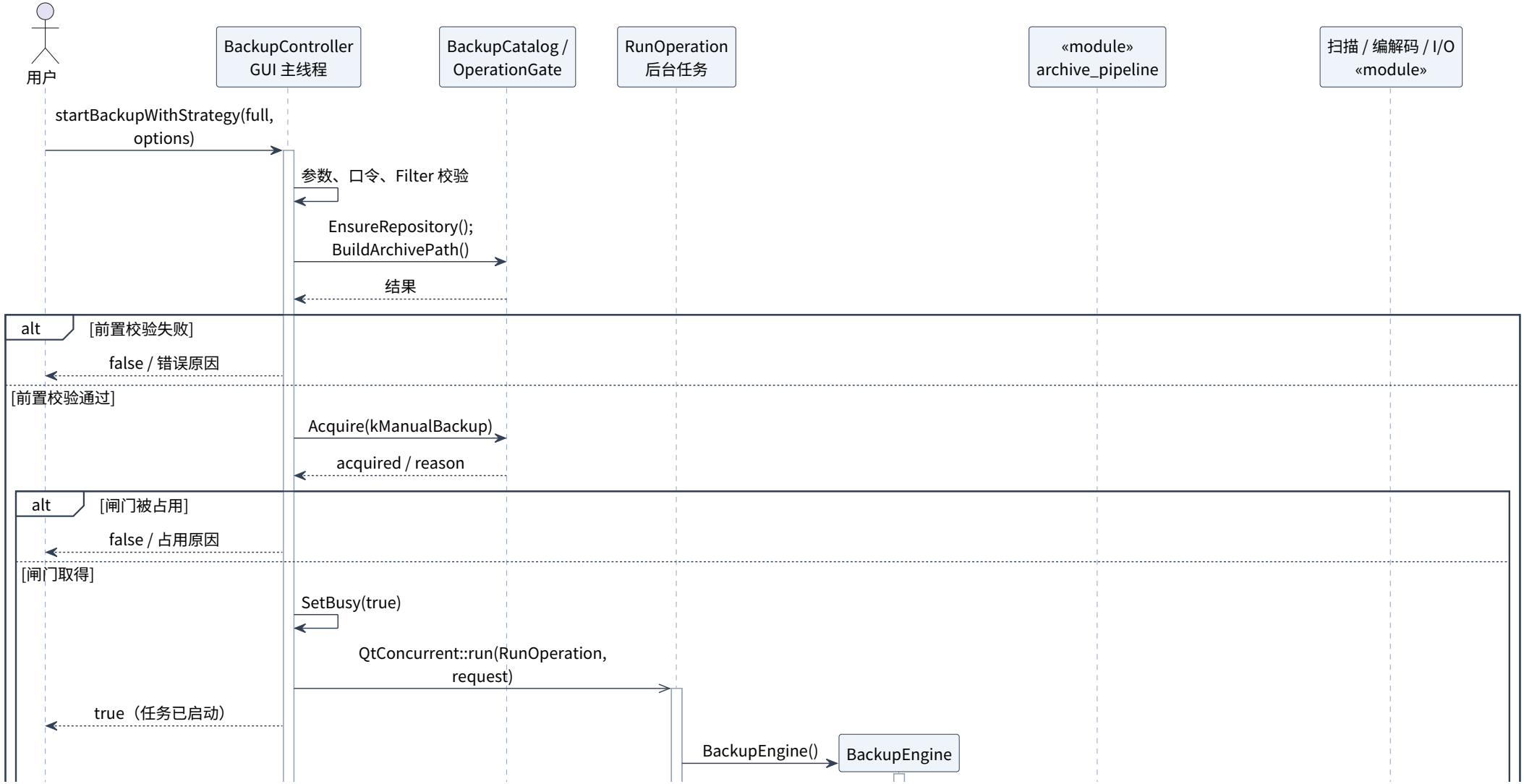
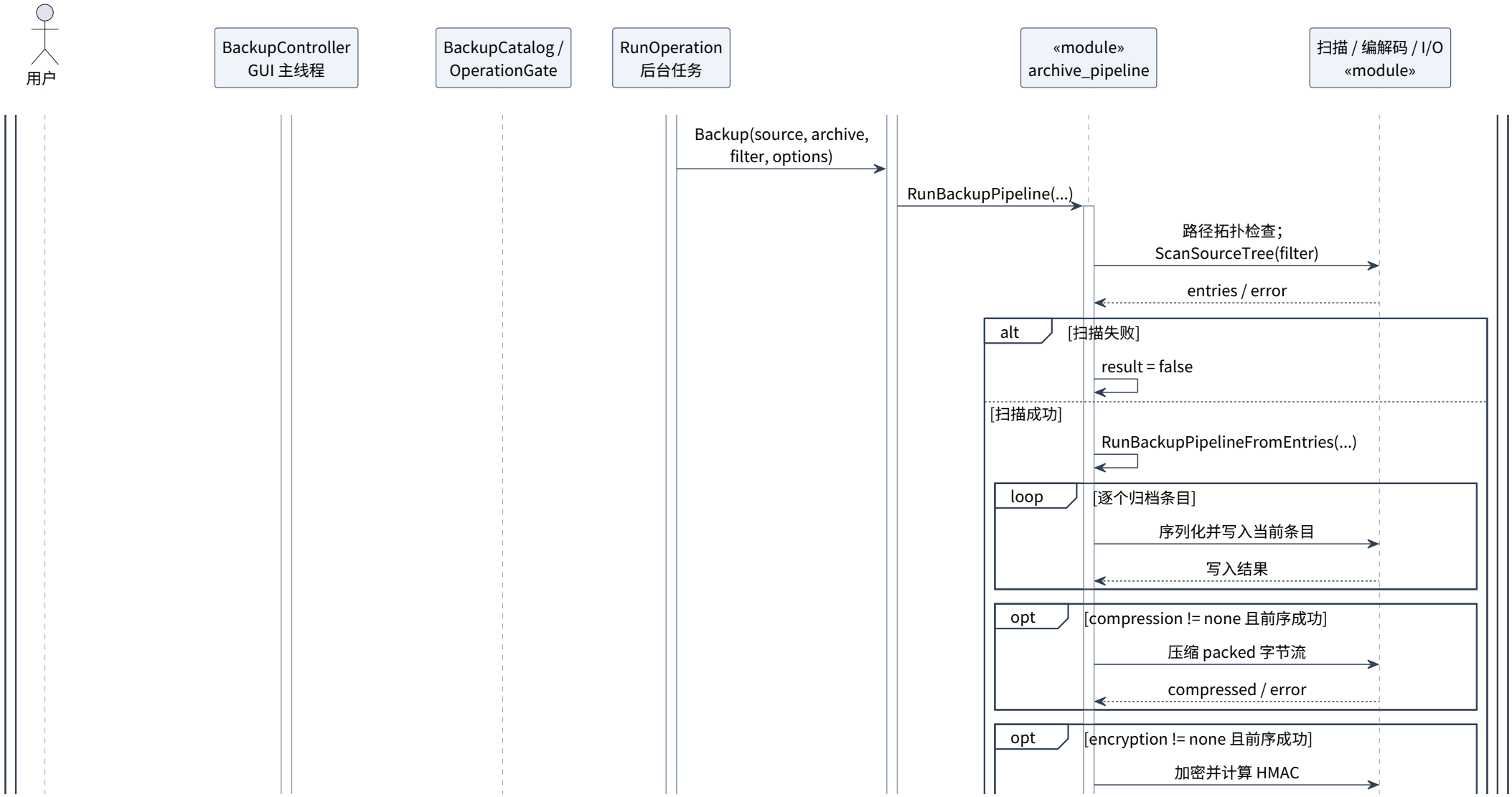
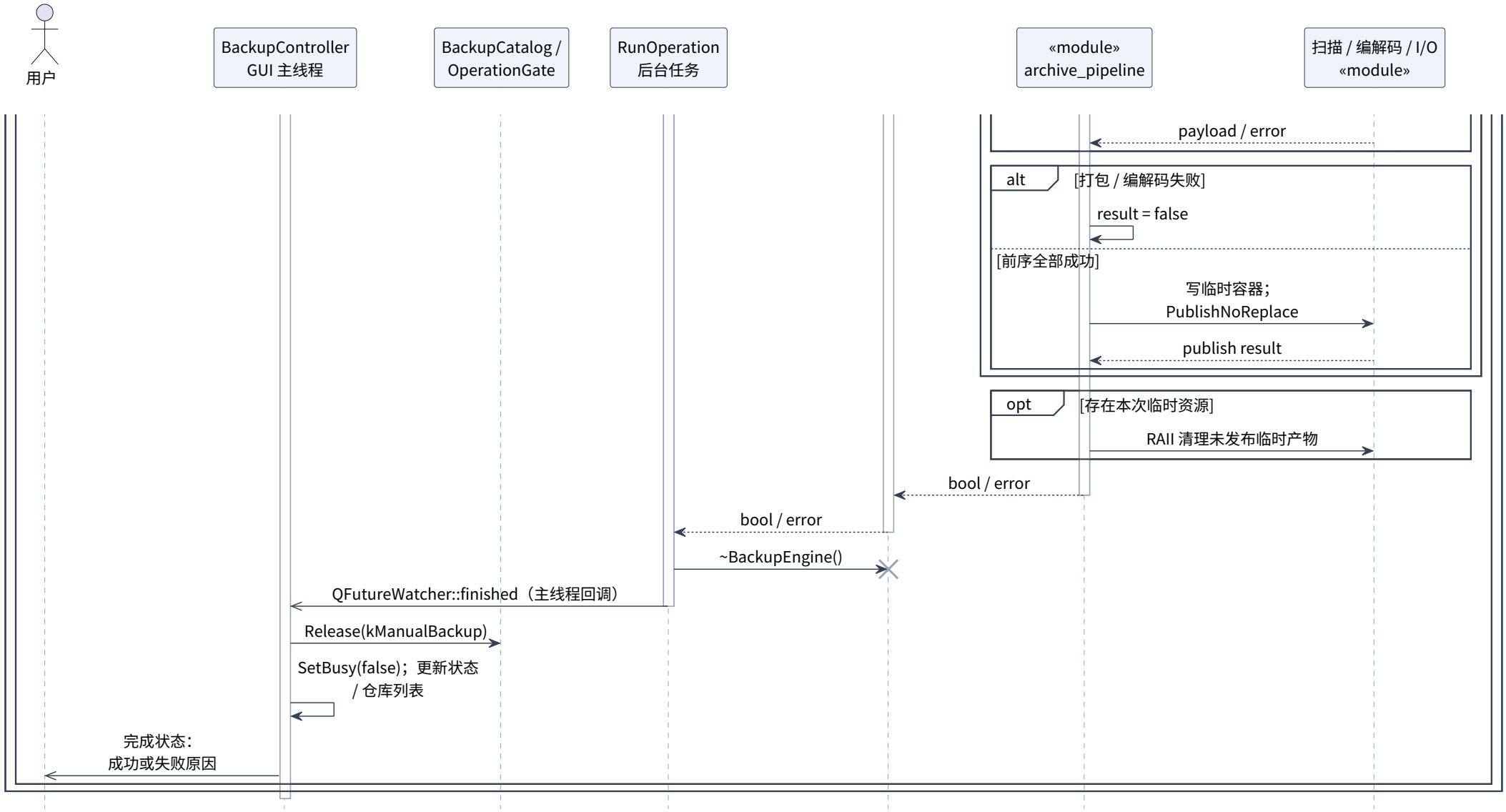

用户
BackupController
GUI 主线程
BackupCatalog /
OperationGate
RunOperation
后台任务
«module»
archive_pipeline
扫描 / 编解码 / I/O
«module»
startBackupWithStrategy(full,
options)
参数、口令、Filter 校验
EnsureRepository();
BuildArchivePath()
结果
alt
[前置校验失败]
false / 错误原因
[前置校验通过]
Acquire(kManualBackup)
acquired / reason
alt
[闸门被占用]
false / 占用原因
[闸门取得]
SetBusy(true)
QtConcurrent::run(RunOperation,
request)
true（任务已启动）
BackupEngine()
BackupEngine


用户
BackupController
GUI 主线程
BackupCatalog /
OperationGate
RunOperation
后台任务
«module»
archive_pipeline
扫描 / 编解码 / I/O
«module»
Backup(source, archive,
filter, options)
RunBackupPipeline(...)
路径拓扑检查;
ScanSourceTree(filter)
entries / error
alt
[扫描失败]
result = false
[扫描成功]
RunBackupPipelineFromEntries(...)
loop
[逐个归档条目]
序列化并写入当前条目
写入结果
opt
[compression != none 且前序成功]
压缩 packed 字节流
compressed / error
opt
[encryption != none 且前序成功]
加密并计算 HMAC


用户
BackupController
GUI 主线程
BackupCatalog /
OperationGate
RunOperation
后台任务
«module»
archive_pipeline
扫描 / 编解码 / I/O
«module»
payload / error
alt
[打包 / 编解码失败]
result = false
[前序全部成功]
写临时容器；
PublishNoReplace
publish result
opt
[存在本次临时资源]
RAII 清理未发布临时产物
bool / error
bool / error
~BackupEngine()
QFutureWatcher::finished（主线程回调）
Release(kManualBackup)
SetBusy(false)；更新状态
/ 仓库列表
完成状态：
成功或失败原因